

CNC Chip Conveyor Timer

Operator's guide · one illuminated button, three programs · board PCB1 rev 1

What it does. The timer presses the conveyor's START and STOP buttons for you, in a loop: run for a while, stop for a while, repeat. While the conveyor stands still the coolant drains off the swarf back into the machine, so the chips reach the bin drier and the coolant stays where it belongs. When the timer is switched off, the START and STOP buttons on the panel work exactly as before.

CONTROLS — ONE BUTTON

A command is a series of quick clicks. Click, click, click... then stop. The command is accepted **0.7 s after the last click** (a beep confirms it).

Clicks	Action
●	Switch ON in the last used program / switch OFF (the conveyor stops)
●●	Program 1
●●●	Program 2
●●●●	Program 3

Choosing a program takes effect at once, timer ON or OFF: the conveyor stops, rests for the program's pause, then the cycle runs. The choice is remembered after power-off. One click switches on with the conveyor starting straight away. The timer always powers up switched **OFF**.

PROGRAMS

Program	Conveyor runs	Conveyor rests
1	90 s	360 s (6 min)
2	90 s	720 s (12 min)
3	90 s	1080 s (18 min)

Each START and STOP press lasts 0.5 s. Longer rest = drier chips, slower removal. Start with program 1; if the chips still come out wet, go to 2 or 3.

Automatic switch-off. After **4 hours** of continuous running the timer stops the conveyor, switches itself off and sounds a 15 s signal. One click starts it again.

WHAT YOU HEAR

Signal	Meaning
Beep, 2 s	Power on — ready after the beep
Beep, 2 s	Program started or changed
Short beep, 0.5 s	Switched off by the button
Long signal, 15 s	4-hour automatic switch-off
Morse SOS ... - - - ...	Settings memory: first power-on (once, normal) — or a memory fault if it repeats at every power-on

WHAT YOU SEE — THE BUTTON LAMP

Lamp	Meaning
Dark	Timer OFF, panel buttons in manual use
Blinks <i>N</i> times, 3 s pause, repeat	Running program <i>N</i>
3 blinks at power-on	Self-test passed
Fast flicker while OFF	Settings memory fault — the timer still works, the program is kept only until power-off. Call service.

GOOD TO KNOW

- Switching off always ends with a STOP press, so the conveyor stops even if it was running at that moment.
- The two relays can never press START and STOP at the same time; every start begins with both released.
- Clicks during the power-on beep are ignored — wait for the lamp to go dark, then click.
- Too many clicks? Anything over four counts as four (program 3).
- A green LED inside the box shows the state for the technician: lit at power-on, one flash every 3 s when OFF, a different rhythm in each phase when running.

Before opening the box switch the machine off. The relays carry the machine's START/STOP circuit; the board runs on 24 V.